

Autonomous Vehicle: Security by Design

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Abstract—Security of (semi)-autonomous vehicles is a growing concern, first, due to the increased exposure of the functionality to potential attackers; second, due to the reliance of functionalities on diverse (semi)-autonomous systems; third, due to the interaction of a single-vehicle with myriads of other smart systems in urban traffic infrastructure. Beyond these technical issues, we argue that the security-by-design principle for smart and complex autonomous systems, such as an Autonomous Vehicle (AV) is poorly understood and rarely practiced. Unlike traditional IT systems, where the risk mitigation techniques and adversarial models are well studied and developed with security design principles such as security perimeter and defense-in-depth, the lack of such a framework for connected autonomous systems is plaguing the design and implementation of a secure AV. We attempt to identify the core issues of securing an AV. This is done methodically by developing a security-by-design framework for AV from the first principles. Subsequently, the technical challenges for AV security are identified.

Index Terms—Autonomous vehicles (AV), security, security by design, cyber-physical systems (CPS), sociotechnical systems.

I. INTRODUCTION

MODERN cars are fitted with a range of autonomy features. To distinguish such cars with varying degrees of autonomy in a consistent manner, SAE International (the Society of Automotive Engineers) proposed 6 levels of autonomy in the standard J3016 [1]. There, Level 0 means no automation, and Level 5 is full automation. From Levels 0 – 2, a human driver monitors the driving environment, whereas, from Levels 3 – 5, the driving system monitors the driving environment. A tabular description of the capabilities associated with different levels of autonomy is presented in Table I.

Understandably, with a higher degree of autonomy, the security risks are also escalated. From Level 3 onwards, it becomes necessary that the car is fitted with an increased number of sensing and communicating devices to be “self-aware.” In the following discussions, unless specifically mentioned, we define Autonomous Vehicles (AV) as modern vehicles with autonomy features at Level 3 and above as specified in Table I.

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TABLE I
AUTONOMOUS VEHICLES: LEVELS OF AUTONOMY
(S - SYSTEM, H - HUMAN)

Level	Automation	Steering Cruise	Environment Monitoring	Fallback Control	Driving Modes
0	None	H	H	H	N/A
1	Supportive	H,S	H	H	Some
2	Partial	S	H	H	Some
3	Conditional	S	S	H	Some
4	High	S	S	H	Some
5	Full	S	S	S	All

II. RELATED LITERATURE

A connected AV is subjected to cyber-attacks through its various network interfaces to the public network infrastructure as well as its direct exposure to the open physical environment [2]–[4]. Reference [5] discusses the effects of security attacks on AV in focus with cooperative driving. Authors [6] use machine-learning models for security for driverless vehicles. Reference [7] proposes cyber-security for smart cars. Reference [8] uses ML schemes to detect spoofing and Denial-of-Service(DoS) attacks. Reference [9] proposes a physical layer security. Reference [10] models safe and stable system via V2X (vehicle-to-everything) interactions ensuring integrity with privacy preserved. Reference [11] discusses on how an attacker can exploit traffic to influence a website. Reference [12] proposes formal methods for security. References [13], [14] security challenges in industry 4.0 and a framework for mitigation.

An attack surface of a system is the collection of the different attack vectors, which is the different points where attackers can make attempts to inject malicious code/data to or extract information from the system to compromise the security control of the AV. Fig. 1 depicts the typical attack surfaces of an AV, such as the ones identified by the authors in [6], [15]–[18], and potential attack sources through which AV security can be threatened. It can be observed that the attack sources are typically external agent/event or even an internal component with malicious intent that attempts to compromise the expected autonomy functionality of the AV. For example, the Bluetooth interface of the AV shown in Fig. 1 can be considered a potential attack surface that can be compromised through plugging malicious devices (attack source) into this communication channel.

There has been a notable body of work on the perceived threats, and the related countermeasures, of AV security [3], [51]–[53] as well as IoT [28], [31]. References [25], [27],

TABLE II

RELATED WORK ON SECURITY ATTACKS, CHALLENGES AND MITIGATION STRATEGIES IN IT SYSTEMS, IoT SYSTEMS, CYBER-PHYSICAL SYSTEMS, AUTOMOTIVE WITH DRIVER (GENERAL VEHICLE / CAR), AUTONOMOUS (GENERAL VEHICLE / CAR)

	Mitigation	Challenges	Attacks	Security and Safety
IT	blockchain [19] risk assessment[20] cost analysis [21] watermarking [11] score system for events [22]		network traffic analysis [11] black-box malware [23] digital control systems [24]	[25] [26] building automation [27]
IoT	vehicular security [28] risk assessment [29]		DoS [30]	[31] [32]
CPS	ML schemes [8] formal methods [12] FMVEA-CHASSIS [33]	industry 4.0 [13] [14]	DoS, replay, deception [34] [8] network attack [2] interactions within CPS components [12]	[33]
Driver (General:G/Car:C)	C - firewall [35] C - neural network based CAN defence [36] G - authentication for CAN [37] [38] [39] G - co-design for secure updates [40] G - cloud-assisted defence for malware [41]	C - [7]	C - DoS [35] G - CAN based malware [42] [43] [44] G - Infotainment system [45] G - replay attacks [46] G - remote attacks [47]	G - [48] G - VANETs [49]
Driverless (General:G/Car:C)	ML models [6]	G - [6] [16] C - [17] [18]	G - [3] C - [5] [4]	G - [50] C - [10]

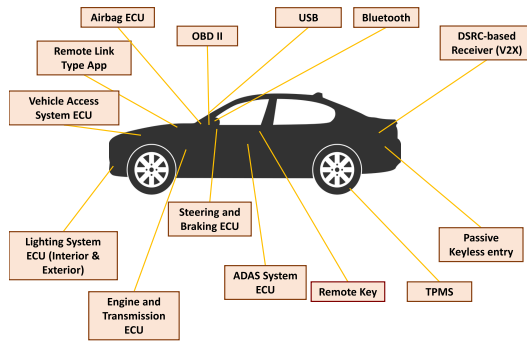


Fig. 1. Potential attack sources and surfaces in cars [15].

[32], [33], [48] argued for a combined safety and security approach. Table II consolidates the related works on security attacks, challenges, and their mitigation strategies for a wide variety of systems: IT, IoT, CPS, driver-driverless (vehicle/car). According to [54], there is 6× increase cyber-attacks in the connected automotive industry, i.e., from 2010 to 2018, there are a total of 170 reported incidents and 60 alone in 2018. 21% of attacks are long-range attacks, where the vehicle is accessed by remote server attack, 8% are keyless entry attacks, 5% are physical attacks (by manipulating CAN bus data), 8% of attacks are by mobile applications (where the attackers breach the app). In total, over 91% are wireless attacks. This clearly shows the need and urgency for “Security by Design.”

In this article, we discuss a more recent trend, AV as a CPS. We argue that these threats can be categorized according to the generic attack models of Cyber-Physical Systems (CPS), i.e., originating from the attack surface of an exposed CPS component. In the next section, we briefly review the security issues of CPS as well as the security of AV as a specific kind of CPS; and particularly emphasize on attacks that have been demonstrated in a practical setting.

III. SECURITY ISSUES OF AUTONOMOUS VEHICLE

In this section, we first review the security issues of CPS before pinpointing the key considerations and new challenges that are specific to AV security.

A. CPS Security

An autonomous vehicle can be considered a specific type of Cyber-Physical System (CPS) and also a kind of Internet-of-Things (IoT) system. Cyber-Physical-Systems are complex, heterogeneous distributed systems, typically consisting of a large number of sensors and actuators connected to a pool of computing nodes. With the fusion of sensors, computing nodes, and actuators, which are connected through various means of communications, CPS aims to perceive and understand changes in the physical environment, analyze the impacts of such changes to the operation of the CPS, and make intelligent decisions to respond to the changes by issuing commands to control physical objects in the system; thereby influencing the physical environment in an autonomous way [55]. As illustrated in Fig. 2, the connections between actuation and sensing through the physical environment, and between sensors and actuators through one or multiple (distributed) computing or intelligent control node(s), form a feedback loop which aims at achieving the desired objective or steady-state. As such, a CPS acts either with full autonomy or at least provides support for a human-in-the-loop mechanism as part of some semi-autonomous control functions. This distributed closed-loop process allows CPS to remotely influence, manage, automate, and control many industrial operations.

Due to the operational nature of CPS in most industrial control processes, CPS are also known as Operational Technology Systems (OT Systems) [56], [57].

The massive adoption of Internet-enabled devices (i.e., IP-enabled sensors and actuators) in CPS systems has thereby blurred the boundary between CPS and Internet-of-Things (IoT). The concept of IoT stems from connected smart devices [58], which may or may not be interacting with physical objects. Hence, there are application scenarios in the classical OT domain that can be conveniently classified both as an IoT system and a CPS system, e.g., a distributed set of sensor nodes to monitor and control the energy usage of a manufacturing plant. Prominent examples of CPS and IoT systems and their corresponding applications [59] include, among others, autonomous vehicles, which is the focus of this work.

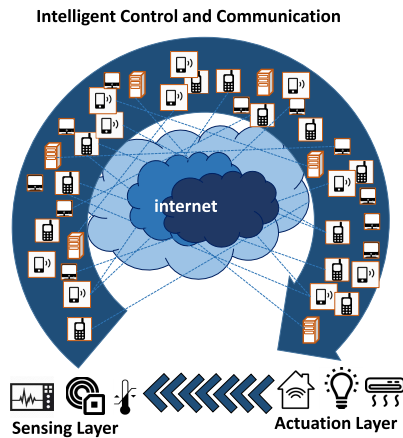


Fig. 2. Interactions between sensor layer and actuator layer.

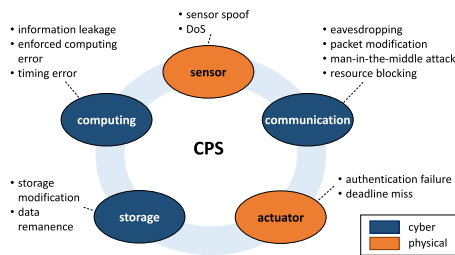


Fig. 3. CPS attacks: Generic model.

As such, attack techniques against a diversity of OT systems have similarities and can be classified as attacks on different CPS components, e.g., communication, storage, actuator, sensor, and computing nodes. A few such attacks are shown in Fig. 3.

However, this naïve and isolated analysis of attacks, in the context of a specific CPS, and the adoption of the corresponding countermeasures are grossly inadequate and misleading for several reasons.

- These generic attack studies [16]–[18], [50] tend to ignore the security objectives of the CPS, which aim to strike a balance between risks, cost and convenience through the adoption of a hybrid of security control measures. Thus, a seemingly insecure mechanism may be operationally acceptable because it is operating within a controlled environment created by other security mechanisms of the system.
- Depending on the prevailing OT security practices, as well as the assumed adversarial model, and it might be unnecessary to account for certain vulnerabilities.
- The generalization of attacks across all CPS typically ignores the roles of Roots of Trust (RoT) and security perimeter modeling, which are the basis of many security-by-design approaches.

In principle, the security-by-design of a CPS is a holistic process that is viewed as a systems engineering discipline [60]. Addressing specific attacks in an isolated and ad-hoc manner cannot help much in security design practices. For these reasons, this paper refers to the above classification and

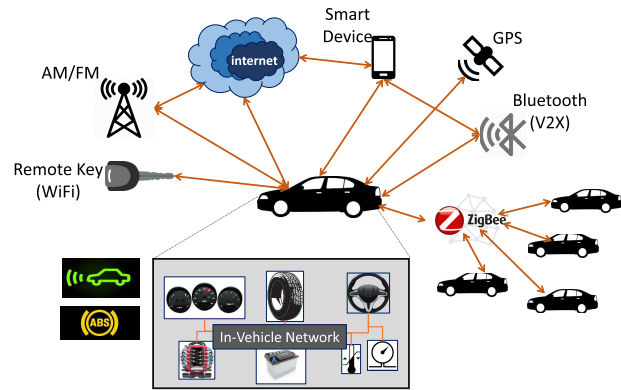


Fig. 4. Exemplary networked vehicle.

enumeration of attacks, as done predominantly in the current literature, as *generic attack* studies in that they tend to study localized attacks in a generic setting of CPS. This trend of generic attack studies is exacerbated by the fact that there is no well-defined security standard that aligns with the road-safety standards for the AV.

In contrast, we emphasize on the security-by-design of AV as a system, indeed a cyber-physical system. In the context of the specific vulnerabilities of AV with different degrees of autonomy, the technological challenges to safeguard AV are derived directly from the underlying safety objectives.

B. AV Security Issues

Fig. 4 depicts the networking capabilities of a typical AV. The car interacts with the outside world with vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication links. The V2I and V2V links provide the AV with, for example, traffic-status information from traffic management infrastructure or navigation-related information received from other AVs on the road. These connection interfaces represent attack surfaces that the adversaries will aim to exploit to obtain unauthorized access to the AV [5]. Thus, it is of paramount importance that these V2I and V2V connections are mutually authenticated and the payload is suitably protected from unauthorized disclosure and unauthorized modification.

Furthermore, in the case of V2V connections, due to the more complicated security scenario, the V2V links are required to operate in a highly reliable and resilient manner.

On the other hand, the internal system of an AV is installed with a multitude of independent and overlapping CPS systems, such as adaptive cruise control, anti-lock braking system, assistive parking system. Depending on the level of autonomy, more functionalities are helmed by the internal CPS controllers, with varying degree of human intervention. These CPS systems are supported by actuators and sensors, such as tyre pressure sensor, crankshaft position sensor, light sensor and obstacle sensors. Today, intra-vehicle communications for helping such CPS controls happen mostly through wired connections known as the Control Area Network (CAN). The intra-vehicle communication protocols typically follow the CAN bus standard or the more recent FlexRay standard.

The security design of AV requires that the intra-vehicle networks [61], [62] are rigorously protected and that access from outside of the vehicle, if applicable (e.g., for maintenance and services of the AV), be strictly controlled. It is evident from previous generic attack studies of AV security that, without proper security designs, even communications through these intra-vehicle wirelines standards are potentially susceptible to security breaches [53], [63]. Specific studies of the intra-vehicle network security have been undertaken in [42]–[44], [52], which is, however, decoupled from the holistic view of the security-by-design principles.

In addition, connections to the internet enable the AV to transmit operational data to the car manufacturer. As a critical system from the perspective of autonomy functions, it is crucial that the communication link between the AV and its manufacturer be authenticated and the payload be appropriately protected in accordance with the nature of the operational data being transmitted from the AV to the manufacturer.

IV. SECURITY-BY-DESIGN

Security designs of business or industrial systems always start from the security objectives of the systems which have to take into consideration the management and people aspects of the systems [64]. Indeed, in real-world situations, security requirements/objectives always start from people e.g., defining business objectives of a system by the business owner, making risk-taking decisions is part and parcel of any business venture. Security is meaningful only when people understand what is at stake. In this connection, the consideration is not only in terms of the cost of security attacks but also the cost of implementing security controls as well as the opportunity cost of limiting business operations due to risk-averse designs. As such, system security is modeled as a socio-technical system in which technical design decisions are heavily influenced by people's factors.

In the context of AV, the prime security objective is inevitably the resilience of safety features of the autonomy functions. As human lives are at stake, it is of utmost importance that the safety of AV is not compromised in the face of attacks [4] on the autonomy functions. Ultimately, AV demands the highest level of security measures to guard against attacks of safety-related features. This is especially so for CPS responsible for collision detection, cruise control and anti-lock braking, etc., because any compromise of such features is likely to result in injuries and even loss of life.

When studying the cybersecurity of socio-technical systems, it is useful to identify the steps and processes where people are involved in the security consideration of the system life-cycle. The people/social/environment factors are typically some of the major sources of vulnerabilities [34], which vary from system to system. A more detailed discussion on socio-technical issues related to security is available at [65], [66].

A general overview of the security management framework for a socio-technical system is presented in Fig. 5 has three faces - front, side, and top faces. Front and the side face of the listed layers map one-to-one from manual to AV. However, the top face is different for an AV and phases from the top face correspond to {transactional security → V2X/V2I},

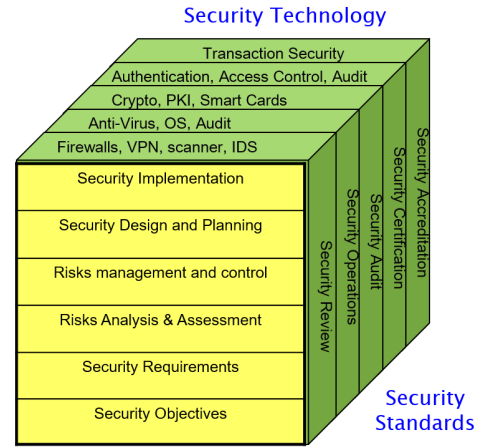


Fig. 5. Layers of security management.

{Authentication/Access Control/ Audit → AV component}, {Crypto/PKI/Smart Cards → In-Vehicle}, {Anti Virus → Firmware protection/ Middleware}, and {Firewall/VPN → Firewall/spoof detector}, where where phases left to → are of manual car and the ones to the right are for AV. In this paper, we discuss the security-by-design of AV as a CPS, and in a holistic manner, by identifying and addressing the security objectives within this socio-technical framework. In the next subsection, important concepts that form the basis of the security design of AV systems are reviewed.

A. Tenets of Security

When designing security for complex systems, it typically involves the following key steps: (1) identify the security objectives and requirements of the system; (2) assess the value and sensitivity of the system to be protected; (3) define security policy in accordance with the security requirements; (4) estimate the capabilities of the adversaries; (5) design control features that commensurate with the sensitivity of the system and the risks, it is exposed to.

Traditionally, standard security policies are guided by the generic Confidentiality, Integrity, and Availability (also known as the CIA triad) requirements at the message/data level. These policies are reinforced with application-specific security requirements and are eventually implemented with a combination of physical and logical control measures including cryptographic primitives and security protocols.

An alternative approach towards security design is to model the potential threats at a system level. For example, Microsoft proposed a threat classification model based on the following six categories, also termed as the STRIDE threat model [67]. More recently, Infineon [68] and Volvo [69] has adopted STRIDE for threat modelling in cars.

- Spoofing of user identity.
- Tampering of stored or communicated data.
- Repudiation, i.e., denying of actions performed where other users cannot prove otherwise.
- Information disclosure or breach of confidentiality.
- Denial of Service (DoS) which makes a network/server unavailable.

- Elevation of privilege for an user to perform unauthorized actions.

In modern CPS and IoT systems, this threat model is discussed together with the notion of attack surfaces. As explained before, an attack surface provides an entry point for an attacker to gain control of or exfiltrate information from the target system. A detailed study of automotive attack surfaces is presented in [53].

B. Adversarial Model

Once the attack surfaces of a system are identified, security designers will need to estimate the likelihood that attacks may happen in real and operational situations. This will mainly depend on (1) the capabilities of the adversaries as well as (2) their interests in investing resources to perform the attacks. The former is typically analyzed by means of some adversarial models while the latter depends on the value and sensitivity of the system to both the system owner and the attacker.

Adversarial model is a formal definition of the attackers' capabilities. Dolev and Yao defined the adversarial model in [70] for proving the properties in an interacting cryptographic system. The adversary, in this model, is capable of hearing, intercepting, initiating and synthesizing any message. However, such a *strong* adversarial model is considered unrealistic; instead, *practical* adversarial models, validated by practical experiments, are used often. Adversarial models have also been used later for network security, e.g., for the security analysis of Onion Ring Routing, which permitted traffic-analysis-resistant and anonymous Internet connections [71]. Adversarial models can also be used for analyzing privacy guarantees, e.g., as has been done in [72], which defined an adversary capable of intercepting the messages and identifying the location of the context-aware system.

In the following, we define different parameters and possible adversarial models for analyzing CPS security. Since the adversarial model is closely tied to the application domain, we use AV as a representative scenario here.

First, the *attack objective*. For a given CPS, the attacker attempts to breach one or more of the security objectives. For example, integrity breaches of an AV would mean that the attacker is potentially able to take over control of the vehicle. As another example, by being able to control communications between the AV and any third-party system, the attacker is capable of violating the confidentiality and integrity of the communication channel, hence can potentially manipulate the safety-related control functions of the AV.

Second, the *communication capability*. The adversarial model is mainly defined by the overall communication capability of the attackers, e.g., to intercept messages transmitted over different communication channels of a CPS, which determines what kind of messages are available to them for analysis and their ability to inject maliciously fabricated messages to the system. In the context of an AV, this could be commands sent through the internal CAN bus of the vehicle or could be messages transmitted as part of some V2X communication.

Third, the *computing capability* of the adversaries. Even after gaining access to the CPS, through breaching the trusted

network, the adversary needs to execute tasks to gain control or damage the CPS. These typically require cryptanalysis of cryptograms, decoding of application protocol messages and reverse-engineering security-critical parameters. The attackers will need to carry out these tasks online or offline in an efficient manner, often requiring considerable computing power.

In the context of AV security, there is no well-defined adversarial model, hence making the work of a security designer harder. Some recent works in the open literature defined adversarial deep learning [73] that can undermine the autonomous driving controls. However, this is not the kind of adversarial model for analyzing the security of the overall AV operations, and is outside the scope of this discussion.

C. Trust Model and Security

As mentioned, security objectives are typically fulfilled by cryptography-enabled control mechanisms that aim to achieve confidentiality, integrity, and authenticity, at least, at the data and message level. This leads to the requirement for key management, which in turn is based on some Trust Model. For example, a Trusted Third Party (TPP) played by an Authentication Server for symmetric key systems or a Certification Authority of public-key systems. Similarly, the use of Web of Trust as in the case of PGP, or Distributed Trust adopted by Blockchain [19], are examples of Trust Models. In the latter, due to the absence of a TPP, one may adhere to the use of consensus protocols and decision mechanisms like the de-centralized trust model of Blockchain.

Trust Model is fundamental to all practical security designs because it lays the security foundation of who one can trust, which in turn allows users to determine what can be trusted. A good security design always starts from a realistic and practical trust model, with other cryptographic mechanisms and security protocols being established and proven based on the trust model.

To illustrate, if the security of a CPS is dependent on information/instruction received from another party, whose trustworthiness is dubious, then the security of the CPS is ill-based. Unfortunately, from experiences, this kind of situation happened very often in practice, most likely, due to the retrofitting approach to the security of a lot of existing CPS. For example, a lot of sensitive online transactions have been hacked because of the underlying key management mechanisms, which are assumed to be trusted, could be based on unverified open-source codes wherein security loopholes are commonplaces. *If the approach of security-by-design is adopted, the underlying trust model will need to be explicitly defined and the basis of such trust will have to be verified and assured.*

D. Trust Infrastructure and Security Mechanisms

In the security design of AV, the communication connectivity infrastructure is of prime importance; besides, it is imperative to support the security administration of trustworthy IoT devices such as key, identity and privilege management of these devices. In a startling difference from wired networks, for wireless and mobile ad-hoc networks, it often admits new

members, and therefore needs to establish secure communication channels frequently. In the following, the distinguishing features of a secure and trustworthy network management infrastructure are presented:

- *Infrastructure for trust management*: A prime use case scenario for IoT devices are ad-hoc sensor networks, which find applications in V2V and V2I communications [74], [75], for example. In such networks, control mechanisms for admitting new nodes and detecting malicious nodes [76] are important prerequisites for maintaining security policies intact. There have been ample studies on the key management protocols for wireless sensor networks, e.g., via key pre-distribution [77], identity-based encryption [78], [79], certification authorities and key exchange protocols. In general, these studies fall under the general theme of trust management [80], which is particularly challenging for low-end devices due to the performance overhead that secure key storage or dynamic code attestation incurs.
- *Secure routing protocol*: IoT systems rely critically on static/dynamic routing protocols, which may be subjected to diverse forms of attacks [81]. Typical countermeasures for routing protocol attacks depend on, firstly, a trusted base station that enables authentication and encryption; secondly, multipath routing and, thirdly, secure geographic routing protocols [82].
- *Heterogeneous network integration*: IoT networks are usually associated with a heterogeneous mix of wireless communication systems, each of which comes with its security protocols [56]. Their interoperability may require the conversion of data formats, which is difficult to undertake without partial knowledge of the message payload. Furthermore, the possible presence of, and often undetected, diverse information channels remains a constant threat [83] that needs to be addressed by secure and trustworthy network management means.
- *Secured Resources*: The computing and storage blocks of a CPS can be considered secure, e.g., as in a Trusted Execution Environment (TEE). The resources under this category need to be assured by only accepting authenticated and signed boot process. Secure storages can be designed by preventing known attacks like data remanence, and further opting for encrypted and authenticated data storage [37] [38] [39].
- *Trusted Identification*: For a CPS resource to be included in the trusted network, it must participate in a robust identification protocol, or be certified by a centralized certification authority. This works in similar principles as in wireless sensor networks, where trust management [80] becomes challenging for low-end devices. For such a resource, trust anchoring can be done using Physically Unclonable Functions (PUFs). Typical adversarial models against these are hardware Trojan elements introduced in the manufacturing/repairing process.

V. SECURITY-BY-DESIGN FOR AUTONOMOUS VEHICLES

In this section, we describe the security assumptions, requirements, threats and control measures of AV by adopting

the security-by-design approach. Specifically, we describe the key steps of security-by-design in the context of AV by viewing it as a socio-technical system. As explained, when designing security for complex systems such as the AV, it typically involves the following key steps: (1) identify the security objectives and requirements of the system; (2) assess the value and sensitivity of the system to be protected; (3) define security policy in accordance with the security requirements; (4) estimate the capabilities of the adversaries; (5) design control features that commensurate with the sensitivity of the system and the risks it is exposed to.

In order to commence the security-by-design process within a sound framework, it is crucial to establish the operation model of the AV so that security objectives and requirements may be analyzed holistically and systematically.

A. AV Operation Model

For simplicity, the following *basic operation model* of an AV is assumed.

- *Communication*: AV periodically sends operation logs to the manufacturer to allow life cycle management and maintenance.
- *Communication*: AV has wireless or wired interfaces to support firmware/software update/upgrade at maintenance and service workshop.
- *Communication*: AV supports communication with some traffic management system infrastructure for traffic flow control as well as a remote control in case of emergency.
- *Sensing*: AV is equipped with a variety of sensors to sense the physical environment and detect collisions.
- *Decision*: AV has several navigation-related control functions that allow Level-5 autonomy, i.e., it requires real-time updating of travel routes, enables intelligent route planning and automatic steering in accordance with road conditions.
- *Decision*: AV has a number of safety-related control functions that allow Level-5 autonomy, i.e., it enables automatic steering, speed regulation, and braking in accordance with road conditions.

B. Security Objectives and Requirements of AV

Based on this operation model, we identify the basic security objectives of an AV as follows:

- *Integrity* of remote control functions of the AV (possibly as an emergency operation from the traffic management system) so that no attacker can take over control of the AV by tampering the remote control system.
- *Integrity* of the sensor systems so that navigation and safety-related control features will not be meddled with by attackers through tampering the sensor data.
- *Integrity* of the safety-related control operations such as braking and speed control is performed in accordance with the sensed road conditions or from remote control instructions.
- *Integrity* of the navigation-related control operations such as steering, braking, and speed control is performed

following the sensed road conditions or from the pre-programming route path.

- *Confidentiality* of communications between AV and traffic management system so that safety-related control parameters will not be disclosed to unauthorized parties who may potentially develop further exploits to the AV.
- *Confidentiality* and *integrity* of communications between AV and its manufacturer so that robustness of the life cycle management and maintenance of the AV can be assured.
- *Integrity* and *authenticity* of communications between AV and maintenance workshop so that software patches and updates [40] to the AV can be performed with high assurance.
- *Confidentiality* of cryptographic keying materials stored inside the AV is ensured so that attackers cannot bypass higher-level security mechanisms by siphoning the cryptographic keys.

C. Safety Standards for AV

AV should be modeled as a socio-technical system where safety is of utmost importance because human lives are at stake. For AV, safety-criticality directly leads to the criticality of cybersecurity, and not the other way around. Hence, the AV security objectives are to be based on, and derived from, the relevant AV safety standards. The current safety standards for AV are presented in the following.

- ISO 26262 [84]: This standard is derived from IEC 61508 [85], which was developed for all electrical/electronic safety-related systems. ISO 26262 is specifically targeted for automotive safety. This standard provides a safety lifecycle throughout the phases like management, development, production, operation, service, and decommissioning. ISO 26262 also defines the Automotive Safety Integrity Level (ASIL). ASIL includes Severity classification (S0 – S3), Exposure classification (E0 – E4) and Controllability classification (C0 – C3) to quantify the severity of an injury, probability of occurrence and controllability of the situation, respectively. ASIL is expressed as follows.

$$ASIL = Severity \times Exposure \times Controllability$$

where a higher level of ASIL indicates a more grievous situation. In the context of AV, it can be noted that the controllability level is extremely high for level 3 upwards. To assess the ASIL, one can adopt techniques such as Hazard Analysis and Risk Assessment (HARA), Fault Tree Analysis (FTA), and Failure Mode and Effects Analysis (FMEA).

- ISO PAS 21448 [86]: This standard is referred to as Safety Of The Intended Functionality (SOTIF) which covers reasonably foreseeable misuse by a person or a misuse which could lead to a hazardous event. In other words, this is applied where situational awareness is derived from processing algorithms and sophisticated sensors (e.g., ADAS, emergency braking systems) where safety is critical. On the contrary, this document excludes

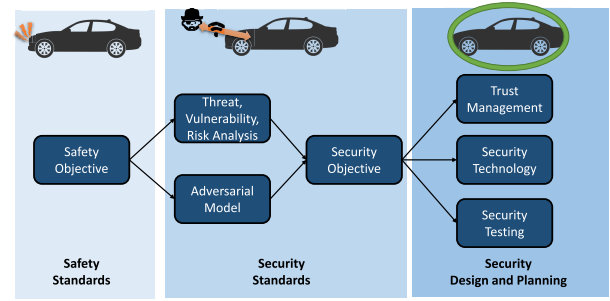


Fig. 6. AV security design flow.

the hazards directly caused by the technology, Intentional alteration/ abuse, and faults covered by ISO 26262.

- SAE J3061 [1]: Recognising the specific need for a standard in the wake of cybersecurity incidents for automotive, the Society of Automotive Engineers (SAE) decided to move together with ISO to define the standard J3061. While it is based on the ISO 26262, it identifies the growing threat landscape and tries to establish awareness and a common terminology across the AV supply chain. It establishes the terminology of threat (malicious attacker), vulnerability (unguarded gateway) and risk (likelihood of attack). Most importantly, J3061 delineates the scope of cybersecurity by stating that the cybersecurity-critical system is not necessarily safety-critical; however, the reverse is true. It also emphasizes the distinction between system safety (fault/accident) and system cybersecurity (purposeful attack).

Fig. 6 illustrates our approach for analyzing and identifying the AV security objectives and requirements based on the relevant AV safety and security policies and standards. In particular, manual vehicle has just first column from Fig. 6, i.e. safety standards. AV has both first and second columns along with security technology from the third column and fully autonomous vehicle has all three columns. Trust Management and Security Testing are specific to fully autonomous because the system by itself has to judge the synergy of data from sensors and then test itself for foreseeable security breaches.

D. Adversarial Models for AV Security

Given the security objectives and requirements as well as the system security model for AV, we now consider the assumed capabilities of the attacks; that is the adversarial models for AV security. As discussed, the purpose of this effort is to estimate the capabilities of the adversaries so that control features can be designed to commensurate with the sensitivity of the system and the risks it is exposed to. The following adversarial capabilities are defined for AV, which are applied for the security analysis for this manuscript.

Property 1: An adversary is capable of intercepting and tampering all inter-vehicle and intra-vehicle communication.

Property 2: An adversary is capable of introducing malicious nodes in the inter-vehicle and intra-vehicle communication network.

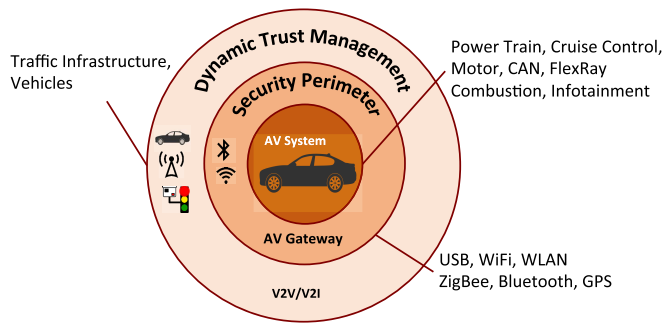


Fig. 7. AV security by design.

These adversarial models are practical and have been demonstrated in the context of several attacks.

E. System Security Model for AV

Though the aforementioned standards can act as the guidelines, to begin with, yet there is a lack of consistency in the approaches for identifying attack surfaces, threat identification, and risk assessment [20]. Consequently, there is no single standard applicable to the AV security as a CPS given its complex overlap across multiple technology domains like wireless communication, electronics [84], mechanical systems and software development practices [87].

Fig. 7 presents a high-level model for security-by-design of AV. In this model, we define the attack surfaces of the AV into three layers: (1) the core layer which is defined by the physical enclosure of the AV; (2) the interface layer, or AV gateway layer, which is characterized by the collection of connectivity interfaces between the AV and the external world; (3) the infrastructure layer which is composed of all the infrastructure and backend modules which are trusted by and connected to the AV. The security issues are distributed over these three layers, first, the AV system; second, the AV gateway; and third, the V2V/V2I communications [10].

The core layer, being defined by the physical enclosure of the AV, enforces its perimeter through physical security. The AV gateway layer consists of diverse forms of communication links, including Bluetooth, WiFi, ZigBee, that help establish and maintain communication with external vehicles and infrastructures. This layer defines a network security perimeter, which is enforced by network access control mechanisms to guard against unauthorized access to the internal functionalities of the AV. This layer relies on the assumed physical security of the core layer so that any damage done to this vehicle within this layer through physical means is not considered an AV security hazard due to cyber attacks.

The outermost layer includes all external systems that are interacting with the AV, which includes vehicles, traffic infrastructures and cloud-based navigation service provider, for example. Note that not all of these external systems could be upfront categorized as a trusted party. Therefore, a dynamic trust management approach has to be undertaken at this layer. A threat arises when a trusted component communicates malicious packets (e.g., malware during software update) or when an untrusted component can bypass the secure gateway

(e.g., by compromising the vehicle sim card). Note that the AV system is not necessarily falling prey to an attacker if the security perimeter is breached. For example, Audi A8 maintains a network layout [88], where the wireless infrastructure is kept away from the internal AV network via a secure gateway. More such defences on vehicle networks include [36], [41], [49]. Thus, a cybersecurity incident may occur but, safety will not be compromised provided the safety-related features are not accessible from the wireless gateway.

VI. AV SECURITY: IMPLEMENTATION, OPERATION AND MANAGEMENT

Typically, the design of security controls starts with the establishment of a security perimeter, followed by a root-of-trust (ROT) definition and the design of trust infrastructure providing the basis for trust management discussed in the following subsections VI-A and VI-B, respectively. This works towards achieving the security objective based on the AV system security model (section V-E). Note that, the security objectives are derived from the safety objective, via TVR analysis and the study of adversarial models.

On the basis of the threat environments due to this perimeter and ROT definitions, we can identify potential security vulnerabilities of AV and, through the identified attack surfaces, we can design the corresponding countermeasures. These countermeasures will be included in the secure AV design framework. We also suggest how these threats/countermeasures could be used as the parameters for AV security testing/auditing in connection to the cybersecurity analysis techniques.

A. Security Perimeter

The purpose of the security perimeter is to divide the AV into segregated security domains with different threat environments. This segregation will have a direct impact on the validity of the trust model, hence the design of the underlying trust infrastructure as well as the security mechanisms. By establishing and accepting a security perimeter, AV design can simplify the distinction between cybersecurity incidents and physical tampering.

More importantly, the use of a security perimeter allows a holistic and systematic approach to security analysis and design for AV. The banking industry, for example, has accumulated vast experiences in the adoption of the security perimeter and defense-in-depth in managing security risks in an organized and controlled manner when their banking systems interface with the open and hostile Cyberspace, to provide Internet banking services.

The three layers defined in section V-E serves as a sound basis for determining the security perimeters for AV. A well-designed security architecture for AV should implement sufficient control mechanisms to reinforce the boundary of each layer so that realistic security assumptions can be made to allow systematic security analysis and design for the layers above it.

To illustrate the importance of the notion of security perimeter in AV, we particularly note that a few related literature emphasizes the sensor spoofing attack as a potential threat

to AV. In our model defined in section V-E, this will not be considered a relevant or applicable threat, because the sensors are embedded within the security perimeter of the core layer. Without open network distribution of the sensors, any attempt to spoof the sensors will require physical intrusion of the AV, e.g., an attacker may damage the rearview camera, thus not considered a cyber attack. On the other hand, a threat is recognized cyber-attacks only if it is due to compromises of the network access control mechanisms at the AV gateway layer or tampered messages from a trusted, but compromised, component.

B. Establishing Roots of Trust

In order to maintain trusted communications with other CPS/IoT devices internal and external to the AV, the ROT must be established. The ROT will serve as the basis for secure key exchange and for key management. Traditionally, a trusted third party (TTP) may be established to enable key exchange and entity authentication. The TTP will play the role of assuring the association of a cryptographic key with the truthful identity of the key owner. Besides, the AV is also assumed to have a secure environment for storing the secret cryptographic key.

In general, though some tamper-resistant hardware may be deployed for storing cryptographic keys in a physically protected environment, the association between the key and the device, identity remains a security challenge. More recently, due to the cost and operational overheads of deploying and administering remote devices such as IoT sensors and actuators, the use of Physically Unclonable Function (PUF) has attracted the attention of IoT security researchers. In essence, PUF allows the device identity to be established directly from the silicon characteristics of the device itself, hence alleviating the device identity administration problem which is fundamental to trust management. Thus, ROT can be achieved by either a PUF-based digital authentication and signature generation; or a public-key protocol to share the key; or having a pre-distributed set of keys.

1) *Trusted Infrastructure*: A trusted infrastructure can be agreed upon during the AV manufacturing and during the AV life cycle. The AV design house needs to provide regular support to the end-users for secure software patches, secure boot and prevention of theft. This can be achieved by marking certain parties as trusted. The smart-phone industry is a prominent example where a number of comprehensive PKI-based trusted infrastructure has been established for enabling device lifecycle management, system software upgrade, application software update as well as for prevention of device theft.

Furthermore, in this context, AV design can be further secured by ensuring secure supply-chain management, which is also a common and recurring theme in the supply chain of Integrated Circuits (IC). Specifically, the planting of Trojan hardware in the AV is a major threat to take into account.

2) *Trust Management*: For V2V/V2I interaction in a given AV environment, the notion of trust has to be rigorously maintained and the level of trust be dynamically established. For example, V2V communication should be based on a

suitable level of trust among the communicating vehicles. Without this, the V2V communication should not be used for supporting critical operations such as safety-related functions in that any malicious (or compromised) participants in the V2V communication will immediately expose a cluster of AVs to a chain-reaction of attacks. Accordingly, the trust in the AV network has to be managed dynamically. On the other hand, for an in-vehicle software update, the patch must be obtained from a trusted source. Though such trusted communication may be established on a need-to basis, the notion of trust must be built on some established ROT.

C. Security of Safety Operations

We now discuss, through exemplary safety scenarios (specifically collision-avoidance and navigation), how some of the key security objectives defined in V-B can be achieved. For all the example cases, we assume the following RoT and security perimeter.

- *Root-of-Trust*: RoT in this scenario includes navigation service provider, which updates and assists on the static/dynamic obstacles in the route of the AV. Furthermore, the integrity of the safety/navigation control components needs to be ensured.
- *Security Perimeter*: We assume it to be AV gateway and the trusted traffic infrastructure for providing navigational services. This security perimeter can be achieved by establishing a secure gateway for the navigation service provider as well as verifying the integrity of the commands.

Table III lists the objectives, RoT, security perimeters, and objectives for vehicle security-by-design in general. A distinction has to be drawn here how these vary from manual to AV. With *manual involvement* in the chain of command (manual and semi-autonomous), there is always an override possible when there is incorrect sensor data (e.g., undetected pedestrian). When the same principle is applied to Table III, manual comes at the boundary of *security perimeter* and ensures the security objectives are met. In the case of AV, both Integrity and Confidentiality has to be managed by AV itself to avoid false corrects.

For the security of safety operations, we exemplify the principles through a few following practical scenarios.

1) *Collision Avoidance*: The attacker can force the AV into collision with another dynamic object (AV, pedestrian) or a static object (road sign).

Security-by-Design: While collision avoidance for AV is a recent concern that requires delving into legal complications apart from the technical challenges, and this is a well-explored topic in the context of vessel [89] and aircraft navigation [90]. Similar techniques with precise location updates and identification of surrounding objects can ensure that the AV does not collide due to its failure. However, from the safety angle, the AV should have a high assurance mechanism for independently detecting a collision, which will have an over-writing privilege to stop the vehicle gracefully if imminent collision is detected. To achieve this, efficient, real-time embedded obstacle sensors will be desirable since the assurance of such kill switch

TABLE III
VEHICLE SECURITY-BY-DESIGN

Safety Objective	RoT	Security Perimeter	Security Objective	
			Integrity	Confidentiality
Collision Avoidance	Trusted Third Parties Key Management	AV Gateway	Sensor, Navigation-system, Kill Switch	Key Management
Navigation	Key Management	Safety-system	Sensor, Navigation-system, GPS Spoof Detection	
Remote Control	Trusted Third Parties Key Management	AV Gateway	Authentication of Remote Control	

mechanism is challenging to attain if it is implemented in software. Worse yet, human lives will be at stake if the collision detection mechanism is based on some AI or analytic techniques whose results tend to be probabilistic.

Here, one can ensure collision avoidance by designing an efficient, real-time, *embedded obstacle sensor-actuator system*. The objective of this system will be to override the car safety/navigation switches and to brake the car and stop the engine, immediately upon the detection of an obstacle that might result in a collision.

To aid this system, the security objective is to ensure - integrity of safety and navigation-related control operations. The specific security measures required are as following.

- *Integrity* of the sensor systems so that navigation and safety-related control features will not interfere with by attackers through tampering the sensor data.
- *Integrity* of the safety-related control operations such as braking and speed control is performed in accordance with the sensed road conditions or from remote control instructions [47].
- *Integrity* of the navigation-related control operations such as steering, braking, and speed control is performed in accordance with the sensed road conditions or from the pre-programming route path.
- *Confidentiality* of cryptographic keying materials stored inside the AV is ensured so that attackers cannot bypass higher-level security mechanisms by siphoning the cryptographic keys.
- *Integrity* of the kill switch that forces the AV to brake/steer in the presence of an unavoidable obstacle, overriding other AV mechanism.

2) *Navigation*: The attacker is capable of mounting an attack through the interface used by a third-party navigation service provider.

Security-by-Design: The attacker can penetrate the internal network of the AV and thus, get access to unprotected components, which are sensitive to the safety and security of the AV. This can be done by either breach-of-trust (by the navigation service provider), or by the software/hardware interface responsible for receiving navigation inputs, such as GPS receivers.

Note that, in this attack, the RoT assumption is violated since we do not necessarily assume the navigation service provider as a trusted party anymore. Here, one can ensure robust navigation by designing a *supplementary GPS system*. The objective of this system will be to provide the car a sense of its position, thus reducing singular dependency on third-party service providers.

To aid this system, the security objective is to ensure - integrity of navigation-related control operations. The specific security measures required are virtually the same as the ones for collision avoidance. Furthermore, one needs to adopt a spoof identification mechanism/supply an independent GPS.

3) *Remote Control*: The attacker is able to achieve control of the car network through a remote control component. Consequently, she/he is capable of taking over the cruise/brake control of the AV, which enables her/him to disrupt the journey of the AV completely.

Security-by-Design: Here, unlike the navigation system, the attacker looks into the remote control access mechanism and uses those gateways for gaining access to the car network. Once the car is accessible, multiple attacks can be mounted, if the AV does not have proper integrity/authenticity/confidentiality checks in place for an intruder. For a remote keyless entry, it is possible to apply some variant of a replay attack citefrancillon2011relay, which records the code (typically from an RF transmitter), and later resends to gain entry to the vehicle.

Note that this attack violates the security perimeter and gain access to the car. To prevent such a scenario, the AV can have an onboard *IoT communication monitor*. This device will maintain all sorts of IoT communications that are related to this particular AV. The device can flag a warning when a set of unusual activities are recorded. The specific security measures required are as follows.

- *Integrity* of remote control functions of the AV, e.g., through biometric identification and/or two-factor authentication.
- *Integrity* of the safety-related control operations such as braking and speed control is performed following the sensed road conditions or from remote control instructions.
- *Confidentiality* of cryptographic keying materials stored inside the AV is ensured so that attackers cannot bypass higher-level security mechanisms by siphoning the cryptographic keys.

D. AV Security Management

While the security-by-design principles attempt to minimize the security risks systematically, security management for AV is necessary for continuous assessment of the security challenges during the entire lifetime of an AV. To that effect, we discuss AV security incidents in the following.

1) *Threat Analysis and Risk Assessment (TARA)*: Since the standards outlined earlier only help in assigning a

quantification under a common assumption of threats, risks, and vulnerability. It is important to assess these factors and undertake rigorous cybersecurity testing, possibly in an automated manner. It should be noted that the Threat Analysis and Risk Assessment (TARA) is specified as part of SAE J3061, which, however, leaves the determination of the risk level and choice of TARA method to the specific organization. To that end, there are few prominent methods listed below, for which a detailed review can be found here [91].

- *Attack Tree Analysis*: A standard method is to base on the Attack Trees, as done for the EVITA Threat and Operability (THROP) analysis [91]. From a functional/feature point of view, the threat is analyzed using the attack trees. Further, with the combination of potential threats, the worst-case scenario is identified and the risk is quantified. The severity classification in EVITA is distributed across safety, privacy, financial and operational verticals.
- *TVRA*: TVRA is a standard approach for Threat, Vulnerability and Risk Assessment for CPS, where the threat is associated with the assets in the system. This was developed for data- and telecommunication applications.
- *Software Vulnerability Analysis* [21], [92]: This is a technique for assessing the vulnerability of software code. The idea of software vulnerability stems from the fact that the development and actual environment of software implementation can differ drastically, more so for AV-like resource-constrained environment and under the influence of a malicious attacker. The Open Web Application Security Project(OWASP) [93] is one such methodology that aims to build secure software projects; this conceptualizes Risk as Impact and Likelihood. These are further decomposed into other factors (e.g., motive, ease of exploit, loss of integrity, and financial damage), which can be weighted based on the importance it has in the business model. Reference [94] uses STRIDE with OWASP for securing connected car ecosystem.

Apart from the above ones, there are further threat analysis approaches based on the STRIDE framework [67], where, first the threat level is determined, followed by the impact level and, eventually, the security level.

2) Security Testing Methods:

- *Penetration Testing*: Penetration testing is commonly performed as part of a security audit, under the setting of a “black box” or a “white box” test subject. In a black-box setting, the system details are unknown to the tester, whereas in the white box set, a powerful adversary is assumed and the system internals are provided to the attacker. The goal of penetration testing to identify the vulnerabilities and assess system security.
- *Red Teaming*: This is a process for detecting network and system vulnerabilities by assuming the role of an attacker, also alternatively termed as ethical hacking.
- *Fuzz Testing*: In fuzz testing, the huge amount of random data is input to the software/system in an attempt to make it crash. The goal is to test coding error and security loopholes.
- *Network Testing*: In this test, the network resilience is tested by bypassing a large number of packets in short bursts.

Further to the stress test, the network configuration and the malicious activities, if any, are tested by performing packet decoding and matching network topology.

Given the diverse kinds of TARA approaches and security testing techniques for secure system design, it is necessary to define and adopt an appropriate methodology for AV security management, especially considering both the OT aspects (repairing/servicing) as well as the IT aspects (software update, malicious nodes).

VII. EXPERIMENTAL STUDIES AND COUNTERMEASURES

In this section, we discuss attacks performed on AV and the experimental setup.

The goal of the setup is to simulate the security attacks and take preventive measures we advocate a secure-by-design approach. Exploiting the vulnerabilities in AV, as discussed in previous sections, we implemented three attacks - *Tyre pressure sensor attack on AV*, *Software update attack on the Infotainment system*, and *Denial-of-Service attack*. Tyre pressure sensor attack - ECU is tricked on false sensor updates, alerting the vehicle to stop. A more severe attack, Software update attack on the Infotainment system [45], not only damages the car but also could lead to fatal injuries, in some cases, death. Denial-of-Service attack, a devastating cyberattack, crashes the system issuing massive number of requests. WIRED reported an attack on the Infotainment system [95] and further recorded driving experience when hacked. Such an attempt by a real hacker is life-threatening. This setup allows many such attacks can easily software simulated in different environments. Experimental attacks have a default city environment (gas-station, zebra crossing, house, etc) with two vehicles - one is AV, the other is Fire Truck(static). For diverse scenarios, Gazebo's [96] features and objects (light-shadow control, traffic lights, wind speed) can be leveraged.

Our experimental system in Fig. 8 includes Gazebo robotic simulator, RViz (3D visualization tool), and Robotic Operating System (ROS). We used Docker [97] to quickly reproduce our results on any ubuntu based system. A custom dockerfile will install all the tools, libraries and dependencies. The experiments are run on Intel Core i7-7500U @ 2.70GHz with 16GB physical memory and 2GB graphical memory (NVIDIA GeForce GT940MX). A base setup is provided by Open Source Robotic Foundation(OSRF) through their GitHub page [98]. However, the available setup is just a skeleton with limited features. To demonstrate the attack, we have extended the code base with proprietary ROS nodes that transmits and receives a message through rostopic (a communication channel in ROS).

A. Tyre Pressure Sensor Attack

What we intended to do? All modern AVs are equipped with many sensor systems to track the condition of the vehicle. One such sensor is *tyre pressure sensor*. The engine control unit (ECU) of the vehicle receives timely updates from the sensor about the status of tyre pressure. In the context of an AV, with lower tyre pressure, the ECU recommends the vehicle to stop. This sensor is embedded in the tyre of the car (internal to

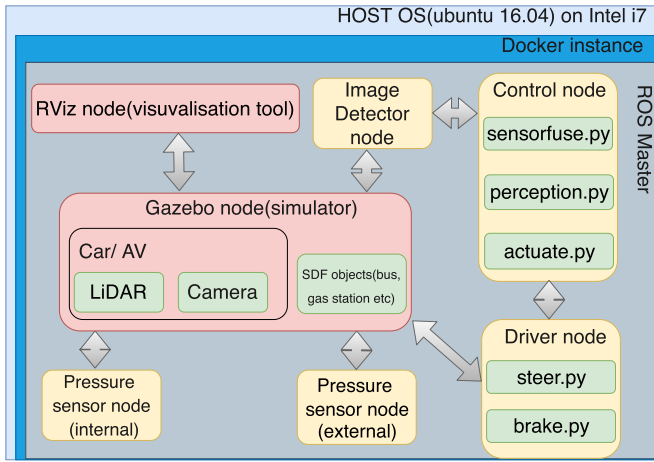


Fig. 8. Simulation system.

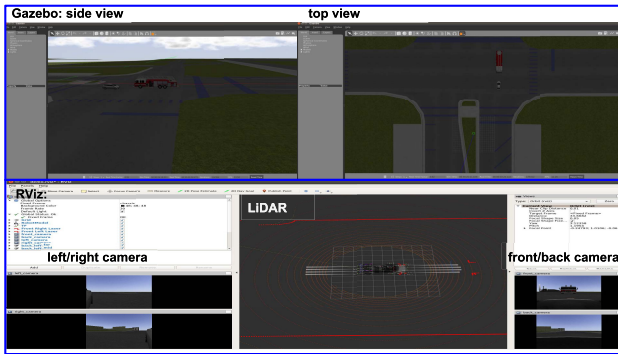


Fig. 9. Gazebo simulator and ROS visualization tool(RViz).

car/ internal sensor). To demonstrate the attack, we considered an external sensor that transmits a false state of tyre. When the vehicle passes in the vicinity of the external sensor, ECU receives a false state of the sensor, and the vehicle stops.

How we did? We have implemented the ECU for the vehicle which takes input from sensors (LiDARs/pressure sensors/image sensors) and controls the vehicle. We then implemented the two sensors (internal/external) which transmit the current state of the tyre (OK: integer '1', NOTOK: integer '0'). The internal sensor is active throughout the x-y grid expect in the region of external sensor ($40 < x - coordinate < 140$). The valid range of the external sensor can be configured in the simulation environment. Implementing ECU/ sensors means extending the codebase by adding extra nodes apart from default ones given by OSRF (Gazebo, RViz, ROSCORE, etc.). So, AV, when passes in-range of external sensor receives false data and stops.

Security-by-Design Approach: Since the external sensor does not violate the security perimeter, the approach we have to adopt is to ensure the integrity of sensor values fed to the ECU. This could be achieved by extracting secure session keys from the RoT and applying hash (key-less) or Message Authentication Code (MAC) for the sensor data.

B. Software Update Attack on Infotainment System

What we intended to do? The infotainment system is one of the most complex systems in modern AV which provides

entertainment and information services to the AV. The infotainment system broadly includes entertainment, security systems, connectivity and driver assistance systems. An attack on that through a false update will affect one of the systems as mentioned earlier.

You only look once (YOLO3) [99] which is an extremely fast and accurate image detection model based on inference from pre-trained weights. A false update of such a crucial application could fail the driver assistance system and crash the AV.

How we did? For this attack, we have implemented four nodes: initialization node, image detector node, control unit node, and driver node. When the simulation starts, the initialization node loads the update/new software release in AV. The labels and weights of YOLO3 are the updates. The YOLO3 algorithm implemented in python and opencv4 libraries are integrated into our simulation. The object detected from YOLO3 is sent to the control unit where the decision making happens based on the hardcoded label. If the recognized label matches the hardcoded label, the control unit node instructs the driver node to stop the AV and prevent the collision. However, in the attack, the weights and labels are corrupted due to a false release of an update. Thereby image detector node label is mismatched with the hardcoded label and leads to a collision.

Security-by-Design Approach: Here, the RoT assumption is violated since, the software update is provided by a third-party service provider, which is untrusted. A chain of trust by mutual authentication of the service providers would eliminate such risks.

C. Denial-of-Service Attack

What we intended to do? The Denial-of-Service (DoS) is a well-known cyberattack - targets webserver by hitting with huge inflow traffic. Last year, GitHub experienced an amplified version of this attack [100], Distributed Denial-of-service, and went down for about 8 minutes. The victims range from content-delivery websites, banks, and news providers [101], [102]. Such an attack from AV perspective can be fatal - unlike information loss/ webserver offline, disrupting the services (cyberattack), a software crash in AV puts human life at risk.

How we did? For this attack, we have implemented trigger node - issue signal to stop, a malicious node - inject traffic to ECU, and a queue for ECU to handle incoming requests from other nodes. In normal operation, AV starts from point-A; continues to move towards point-B, and once it reaches point-B, trigger node signals ECU to stop. However, with malicious node injecting false traffic, the trigger signal can be delayed. The 'delay' is obscure; depending on the severity of the attack, here it is the number of injected packets; in case of mild attack, i.e., when the queue is partially filled, the trigger signal is delayed, but eventually ECU receives signal and it stops at point-C. In the worst case, with constant influx traffic, the queue would always remain full with injected traffic leaving no space for a trigger signal and the AV never stops.

Security-by-Design Approach: Here, the security objective is again to validate the integrity of every control signal. Since such fine-grained integrity monitoring could hurt performance,

one may alternatively opt for periodically checking the authenticity of the nodes through remote attestation.

The above attacks highlight the need for stringent security measures in AV and the direct applicability of security-by-design principles.

VIII. CONCLUSION AND OPEN PROBLEMS

Security of AV is an important issue and should be considered systematically and holistically. In this paper, we argued that the security-by-design principle for AV is poorly understood and rarely practiced. We addressed the issue by modeling an AV as a cyber-physical system and studied the AV security objectives by viewing from the perspective of a socio-technical framework. This was done methodically by developing a security-by-design framework for AV from the first principle.

We derived the security objectives and the necessary control measures from the perspective of the safety requirements of AV. We argued that one of the critical goals of cybersecurity of AV is to ensure that safe operation is resilient in the face of cyber-attacks. Subsequently, the technical challenges and the proposed approaches for AV security were identified and discussed.

Nevertheless, apart from safety assurance, for AV to be adopted as a preferred means for transport, the legal and liability issues behind AV remain a significant challenge. In essence, technical designs and control measures should be developed to enable law-enforcement agencies and judicial officers to determine the liabilities and the parties at fault in the unfortunate situation of car crashes which could lead to loss of lives. The legal and liabilities issues are essential problems that should be addressed as part of the future studies of AV security.

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